Dhole Patil College of Engineering, Pune

Engineering Chemistry

Unit 4 - Fuel

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Fuel

- **Definition**: Fuel is the substance which on combustion produces a large amount of heat.
- **Classification of fuels :**

On the basis of occurrence fuel can be classified into

i)Primary (Natural) fuel- Primary fuel exist naturally eg. Wood, coal, vegetable oils, crude oil etc.

It is further classified as

Solid Fuel : wood, coal etc.

Liquid fuel : Crude oil, vegetable oil etc

Gaseous fuel : Natural gas

- **ii) Secondary (derived) fuel-** These fuels are obtained from primary fuel by processing or they are man made.

- eg 1) Charcoal is obtained from wood by partial combustion of wood.
- 2) Ethyl alcohol is obtained by fermentation of carbohydrates.

- It is further classified as
- **Solid fuel** : Charcoal, coke, nitrocellulose etc.
- **Liquid fuel** : Alcohol, petrol, diesel, biodiesel etc.
- **Gaseous fuel** : Water gas, producer gas etc.

Fuels	Solid fuels	Liquid fuels	Gaseous fuels
Primary (Natural)	Wood, peat, coal	Crude oil	Natural gas
Secondary (Derived)	Coke, charcoal	Diesel, petrol, kerosene	CNG, LNG, oil gas

Characteristics of an ideal fuel

- i) High calorific value
- ii) Moderate ignition point
- iii) controllable velocity of combustion
- iv) Harmless products of combustion
- v) No ash (noncombustible matter)
- vi) Cheap
- Vii) Easy risk-free transportation
- Viii) Smaller space to store
- Ix) Exact air requirement
- X) use in internal combustion engine
- Xi) No moisture
- Xii) No volatile matter

Calorific value

- **Definition**: It is defined as the amount of heat obtained on complete combustion of unit mass of solid or liquid fuel or unit volume of gaseous fuel at STP.
- **Classification :**

There are two types of calorific value

1. **Gross calorific value (GCV) :** It is defined as the total amount of heat obtained on complete combustion of unit mass of solid or liquid fuel or unit volume of a gaseous fuel (STP) on cooling the products of combustion to 15^0c.

- The GCV is called as Higher calorific value (HCV).

- **Net Calorific value:** It is the amount of heat obtained practically on the complete combustion of unit mass of solid or liquid fuel or unit volume of a gaseous fuel (STP) and the products of combustion are allowed to escape with some heat.
- NCV is also called as lower calorific value (LCV)

- **Relation between GCV and NCV**

NCV and GCV are related as

$$\mathrm{G.C.V} = N.C.V + \frac{9\,X\,h\,X\,Latent\ heat\ of\ water}{100}$$

h = Percentage of hydrogen in fuel.

- **Units of Calorific value**

System	Solid/Liquid fuel	Gaseous fuel
CGS	Cal/gm	Cal/lit
MKS	Kcal/kg	Kcal/m^3
SI	Joules/kg	Joules/m^3

- **The interconversion**
- 1 cal/gm = 1kcal/kg = 4.18kj/kg
- (The latent heat of steam is 587cal/gm **or** 2450 kj/kg **or** 587 kcal/kg)

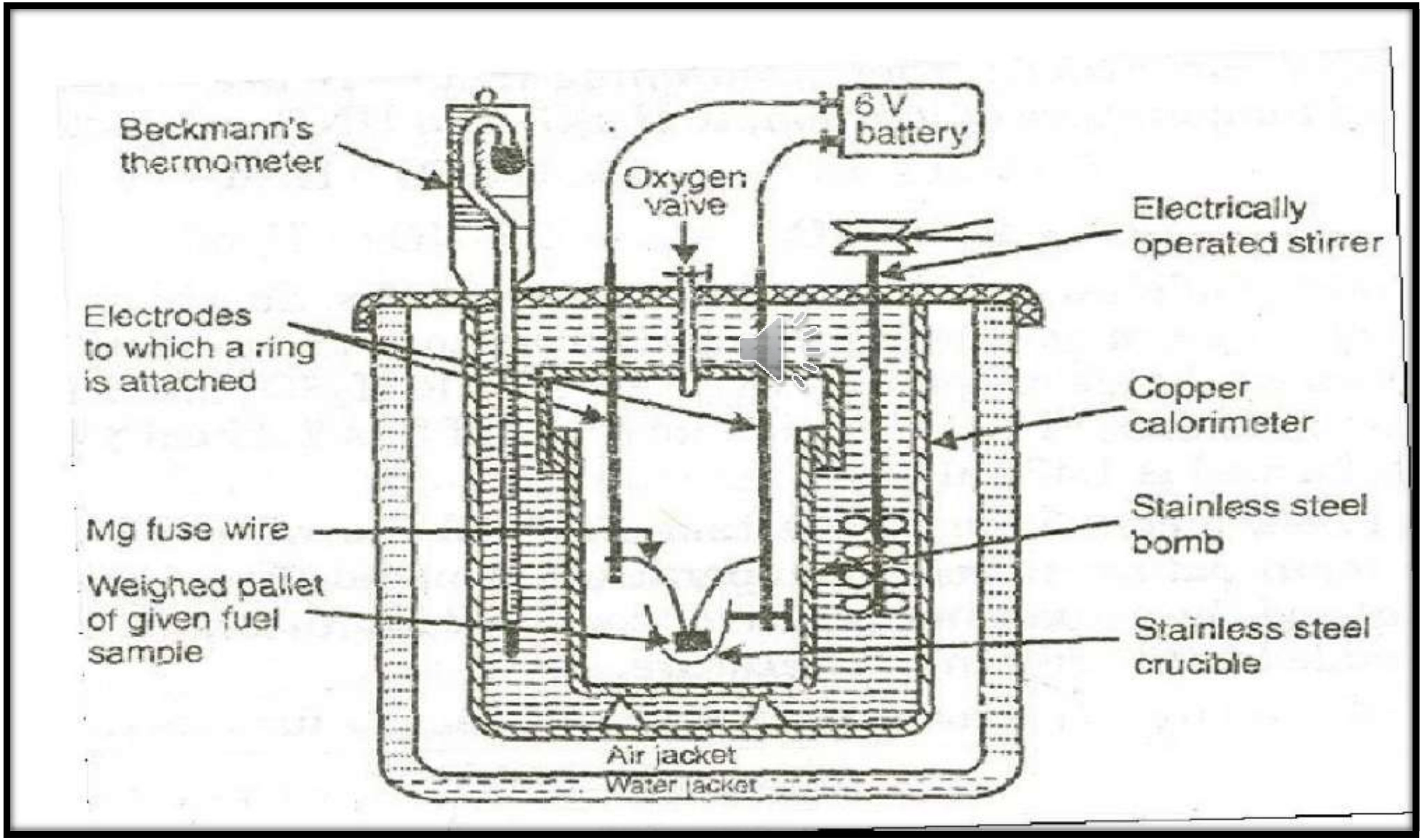
Bomb Calorimeter:

- It is used for determining calorific value of solid & liquid fuels.

- **Principle**:

  The fuel is burnt in calorimeter and the heat produced is transferred to the surrounding water. From this rise in temperature of water, calorific value is calculated.

Bomb Calorimeter: Figure



Construction

It consists of four parts

1. **Bomb Pot:** It is a cylindrical, strong stainless steel pot having a lid. There are two electrodes fitted through the lid and there is an oxygen inlet valve at its centre. The weighed fuel is burnt in the presence of high oxygen pressure.

2. **Calorimeter:** There is a steel or copper calorimeter in which bomb pot is kept. A Beckman thermometer is kept to record the temperature of water.

3. **Water & air jackets:** The calorimeter is surrounded by air and water jacket to avoid heat losses due to combustion.

4. **Accessories:** There is a Oxygen cylinder with pressure gauge to fill oxygen in the Bomb calorimeter. There is also a DC battery of 6 Volts, to start combustion of fuel.

Working of Bomb Calorimeter :-

1. Weigh the sample in crucible. Keep crucible in ring of electrodes. The resistance wires should touch it.
2. Fill bomb with O_2 & place it in calorimeter.
3. Add known amount of water in calorimeter.
4. Keep thermometer in water, start stirrer.
5. Place cover on top. Make electrical connections from battery to electrodes. Note initial temp.
6. Pass current for 5-10 sec to heat wire so that fuel catches fire..
7. Note down max. temp.

Observations:

1. Mass of fuel in gm = X gm

2. Mass of water in calorimeter = W gm

3. Water equivalent of calorimeter = w gm

4. GCV of fuel = L calories/gm

5. Rise in temperature of water = $(t_2 - t_1)$

Calculations:

Heat liberated by burning of fuel = Heat absorbed by water & calorimeter

$$XL = W (t_2 - t_1) + w (t_2 - t_1)$$

$$XL = (W + w) (t_2 - t_1)$$

$$G.C.V. = \frac{(W+w)(t2-t1)}{X} \text{ cal/gm}$$

Corrected Formula

To get more accurate results, following corrections are to be taken into account.

1. **Fuse wire correction (f)**
2. **Acid correction (a)**
3. **Cooling correction (t_c)**

1.Fuse wire correction (f):

Out of the total heat obtained, little heat is given out by fuse wire when the current is passed for 5-10 sec to start the combustion. Hence **Fuse wire correction must be subtracted.**

- **2. Acid correction (a):** The S and N in the fuel are oxidized to H_2SO_4 and HNO_3. Formation of H_2SO_4 and HNO_3 are exothermic reaction and the heat measured includes a small share by these acid formation.

- $S + O_2 \rightarrow SO_2 \xrightarrow{O_2} SO_3 \xrightarrow{H_2O} H_2SO_4 \quad \Delta H = -144$ kcal/mol

- $N + O_2 \rightarrow NO_2 \xrightarrow{O_2, H_2O} HNO_3 \quad \Delta H = -57$ kcal/mol

- **Acid correction must be subtracted.**

- **3. Cooling correction (tc):** If the time taken for water in calorimeter to cool from maximum temperature attained to the room temperature is 't' minutes and average cooling rate is dt/min, then the cooling correction to be added to rise in temperature is t.dt.

- **Cooling correction must be added**.

After taking all corrections into account, the corrected formula for GCV is

$$G.C.V. = \frac{(W + w)(t2 - t1 + tc) - (a + f)}{X}$$

•Neumerical

- **Q.1 Calculate GCV and NCV of a fuel containing 7.5% hydrogen with the following data of Bomb calorimeter experiment. Mass of fuel is 0.928 gm, Water equivalent of calorimeter set- 850 gm, Mass of water in calorimeter-2050 gm, rise in temperature 2.78 ^{0}c.**
- **Solution: Given data**
- X= 0.928 gm, w=850 gm, W=2050 gm t_2-t_1 = 2.78 ^{0}c
- $G.C.V. = \frac{(W+w)(t_2-t_1)}{X}$ Cal/gm or Kcal/kg
- $= \frac{(850+2050)(2.78)}{0.928}$
- = 8687.5 cal/gm or Kcal/kg

- $$\mathrm{G.C.V} = N.C.V + \frac{9\,X\,H\,X\,Latent\ heat\ of\ water}{100}$$

- $$\mathrm{N.C.V} = G.C.V - \frac{9\,X\,H\,X\,Latent\ heat\ of\ water}{100}$$

$$= 8687.5 - \frac{9\,X\,7.5\,X\,587}{100}$$

= 8291.3 Cal/gm or Kcal/kg

- Calculate HCV of 0.75 gm of a fuel containing 80% carbon, when burnt in a Bomb calorimeter increased the temperature of water from 27.3 to 29.7^{0}c. The calorimeter contains 250 grams of water and it's water equivalent is 150 grams.
- Solution:
- **Given data-** Weight of fuel = x = 0.75 gm

  Weight of water = W = 250 gm

  Water equivalent = w = 150 gm
- t1 = 27.3^{0}c, t2 = 29.7^{0}c

- HCV (L) = $\frac{(W+w)(\qquad)}{x\, t\,2-t}$ Cal/gm or Kcal/kg

  $$= \frac{{}^{1}(250+150)(29.7-27.3)}{0.75}$$

  = 1280 cal/gm

- **Q.2 Calculate G.C.V. of a liquid fuel if the data in Bomb calorimeter experiment is**
- **Mass of liquid fuel burnt =1.025gm**
- **Rise in temperature of water = 3.04^0c**
- **Mass of water in calorimeter = 1900 gm**
- **Water equivalent of calorimeter= 770 gm**
- **Cooling correction= 0.07^0c**
- **Acid correction = 43.9 calories**
- **Fuse wire correction = 12.8 calorie.**

- **Solution:**

Given: X=1.025 gm, t_2-t_1 = 3.04 ^{0}c, W = 1900 gm, w = 770 gm, a = 43.9 cal, f = 12.8 cal, tc= 0.07^{0}c

$$G.C.V. = \frac{(W+w)(t2-t1+tc)-(a+f)}{X}$$

$$= \frac{(1900 + 770)(3.04 + 0.07) - (43.9 + 12.8)}{1.025}$$

= 8045.8 cal/gm or Kcal/kg

- 0.60 g of a fuel on complete combustion in a Bomb calorimeter increased the temperature of water in calorimeter from 26.3 ^{0}c to 28.4^{0}c. The calorimeter contains 500 g of water and water equivalent of the calorimeter is 1500 g. The corrections are as follows-

Cooling correction = 0.02^{0}c

Fuse wire correction = 15 cal

Acid correction = 65 cal

If the fuel contains 5% H, Calculate GCV &

NCV of the fuel.

- **<u>Solution</u>**: Given – Weight of fuel burnt = x = 10.60 g
- Weight of water in calorimeter = W = 500 g
- Water equivalent of calorimeter = w = 1500 g
- Initial temperature of water = t_1 = 26.3^{0}c,
- Final temperature of water = t_2 = 28.4^{0}c
- Cooling correction = t_c = 0.02^{0}c

- Fuse wire correction f = 15 cal
- Acid correction = a = 65 cal
- % H in fuel = 5

$$G.C.V. = \frac{(W + w)(t2 - t1 + tc) - (a + f)}{X}$$

$$= \frac{(500 + 1500)(28.4 - 26.3 + 0.02) - (65 + 15)}{0.60}$$

= 6933.33 cal/gm or Kcal/kg

NCV = GCV – 0.09 H x Latent heat

= 6933.33 – 0.09 x 5 x 587

= 6669.18 cal/g

Numerical for practice

1. 0.072 g of a fuel containing 80% Carbon, when burnt in a Bomb calorimeter, increased the temperature of water from 27.3^0c to 29.1^0c. If the calorimeter contains 250 g of water and its water equivalent is 150 g, calculate GCV of the fuel.
2. A Sample of coal containing 5% of H when allowed to undergo combustion in Bomb calorimeter, the following data were obtained.

   Weight of coal burnt = 0.95 g

   Weight of water taken = 700 g

   Water equivalent of Bomb calorimeter = 2000 g

   Rise in temperature = 2.48^0c, Cooling correction = 0.02^0c

   Fuse wire correction = 10 cal

   Acid correction = 60 cal

   Calculate GCV and NCV of coal.

Boy's Calorimeter:

It is used for determining calorific value of gaseous fuels.

Principle:

A gaseous fuel is burnt at a known constant rate in the calorimeter, so that total amount of heat produced is absorbed by circulating water. From the rise in temperature of water calorific value can be calculated.

Construction:

It consists of following parts

1. Gas burner
2. Combustion chamber
3. Thermometers
4. Insulating cover

Boy's Calorimeter:

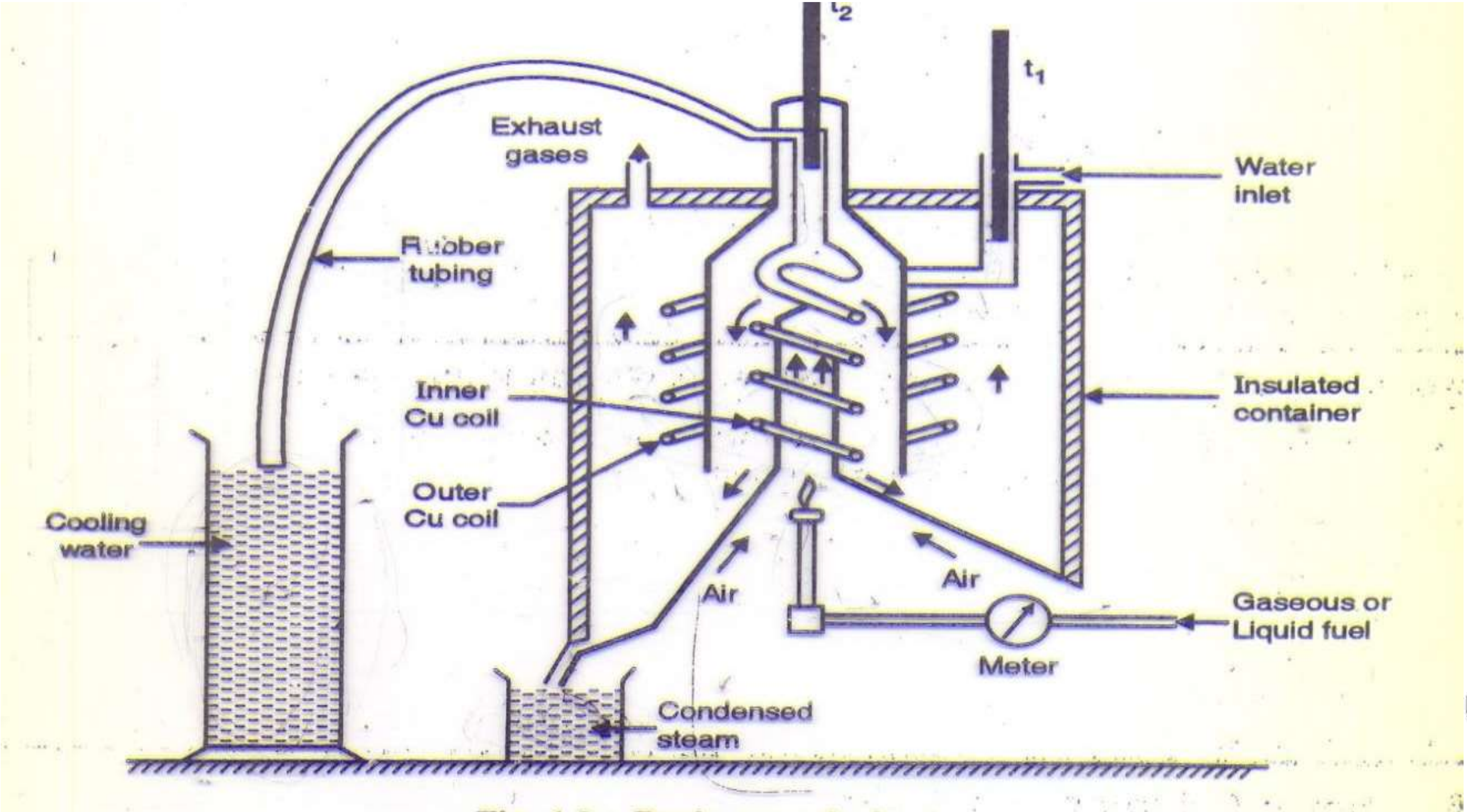


Fig. 1.2 : Boy's gas calorimeter

1.Gas burner: There is a gas burner in which known volume of gas is burnt at a known pressure. The gas is burnt at the rate of 3-4 litre per minute.

2.Combustion chamber: Around the burner there is a combustion chamber which has a copper tubing coiled inside as well as outside of it. Water enters from the top of the outer coil, moves to bottom of chimney and then goes up through the inner coil to the exit at top.

3.Thermometers: There are two thermometers to measure temperatures of inlet water and outlet water.

4.Insulating cover: The assembly is covered with an insulator to detach combustion chamber from atmosphere. There is a hole on top for exhaust gas, water inlet and condensed steam comes out from bottom outlet.

Working :

1. Burn the gas at suitable pressure & adjust the rate of flow such that the temperature of outgoing water remains constant.

2. After steady conditions note down the following observations

Observations :

(a) Volume of gas burnt at given temp. & pressure in certain time period $= V \ m^3$

(b) Quantity of water passed through coil during this period = W

(c) Mass of water condensed during the period = m kg

(d) Rise in temp of water $(t_2 - t_1)^0c$

(e) GCV of the fuel $= L \ Kcal/m^3$

Calculations:

i) Heat liberated by burning of fuel = Heat absorbed by circulating water

$$VL = W(t_2 - t_1)$$

$$G.C.V.\ (L) = \frac{W(t2-t1)}{V} \text{ Kcal/m}^3$$

$$N.C.V.\ (L) = \frac{W(t2-t1)}{V} - \frac{m\ X\ 587}{V} \text{ Kcal/m}^3$$

$$N.C.V.\ (L) = G.C.V - \frac{m\ X\ 587}{V} \text{ Kcal/m}^3$$

Numerical

Q.1 Observations in the Boy's gas calorimeter experiment on a gaseous fuel are given below. Find the G. C.V. and N. C. V. of the fuel, volume of gas burnt (STP) = 0.08 m^3. Mass of cooling water used = 29.5 kg. Rise in temperature of circulating water= 9.1^{0}c. Mass of steam condensed=0.04kg.

Solution: Given: W= 29.5 kg, V= 0.08 m3, (t2-t1) = 9.1^{0}c, m= 0.04 Kg

$$G.C.V.\ (L) = \frac{W(t2-t1)}{V}\ \text{Kcal/m}^3$$

$$= \frac{29.5\ X\ 9.1}{0.08}\ \text{Kcal/m}^3$$

= 3355.6 Kcal/m^3

- $N.C.V.\ (L) = \frac{W(t2-t1)}{V} - \frac{m\ X\ 587}{V}$

$$= \mathrm{G.C.V.} - \frac{m\,X\,587}{V}$$

$$= 3355.6 - \frac{0.04\,X\,587}{0.08}$$

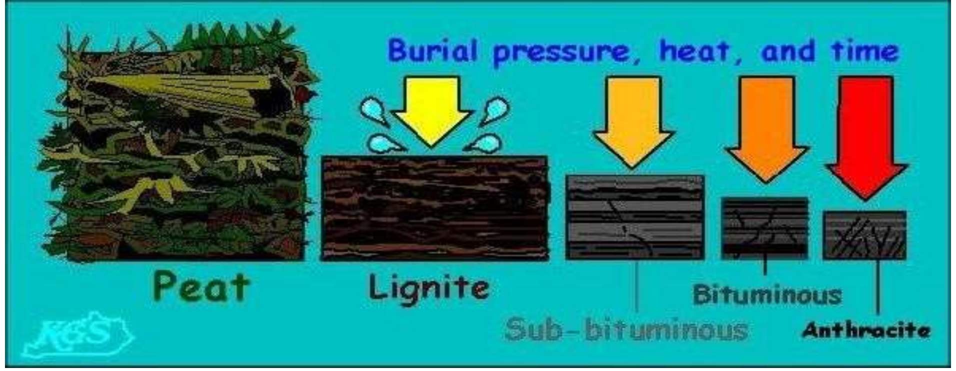
= 3062.1 Kcal/m^3

Numerical for practice

- In Boy's gas calorimeter experiment when 0.1 m^3 of a fuel gas is burnt during which 25 kg of water is circulated. Temperature of incoming water and outgoing water is 20 ^0c & 33 ^0c respectively. Weight of steam condensed is 250 gm. Calculate gross calorific value and net calorific value, if heat liberated in condensing water vapour and cooling condensate is 586 kcal/kg.

- The following observations were noted in the Boy's gas calorimeter experiment-volume of gas burnt at STP=0.1 m^3, mass of cooling water used = 25 kg, rise in temperature of circulating water = 9.1 ^0c. Mass of steam condensed = 0.04 kg. find the GCV & NCV of the fuel.

SOLID FUELS



Coal

- Coal is a highly carbonaceous matter that has been formed as a result of alteration of vegetable matter (e.g., plants) under certain favorable conditions.

- It mainly composed of C, H, N, and O, along with non-combustible inorganic matter

Wood → Peat → Lignite → Bituminous → Anthracite

Changes in Properties from peat to Anthracite are

↑ C %, calorific value, hardness, density, lusture, intensity of black colour

↓ Moisture content, volatile matter, ash, O, N, H and S %

Analysis of Coal

1. **Proximate Analysis:**

   The process of analysis of % **Moisture, Volatile Matter, Ash & Fixed carbon** of coal is called as Proximate analysis.

2. **Ultimate Analysis:**

The process of analysis % of **C, H, N, S, & O** is called as Ultimate analysis.

Proximate analysis:

1. % of Moisture

- **Principle:**

  All moisture in coal escapes on heating coal at 110^0c for 1 hour.

- **Method:** A known weight of powdered and air dried coal sample is taken in a crucible and it is placed in an oven for 1 hr at 110^0c. Then the coal is cooled in a desiccator and weighed out. If the initial weight of the coal is m gms and final weight is m_1 gms . The loss in weight ($m-m_1$) corresponds to moisture in coal.

- **Formula:**

$$Moisture\ \% = \frac{Weight\ of\ moisture}{Weight\ of\ coal} X\ 100$$

$$= \frac{m - \mathrm{m}_1}{w} X\ 100$$

Significance of % Moisture:

- Decrease in calorific value of coal largely, as it does not burn and takes away heat in the form of latent heat.
- It increases the ignition point of coal.
- Hence, a coal with lower moisture % , better is the quality.

2. % of Volatile matter

- **Principle:**

  At 925^0c, coal molecules undergo thermal degradation to produce volatile matter.

- **Method:** Moisture free coal left in the crucible in first experiment (m_1) is covered with a lid loosely. Then it is heated at 925^0c for 7 minutes. The crucible is taken out and cooled in a desiccator. The it is weighed (m_2) gms. The loss in weight (**m_1** - m_2) gms is due to the loss in volatile matter in the m gms of the coal.

- **Formula:** $Volatile\ matter\ \% = \frac{Weight\ of\ V.M}{Weight\ of\ coal} X\ 100$

$$= \frac{m1 - m2}{W} X\ 100$$

Significance Of % Volatile Matter:

- It decreases calorific value of coal.
- It elongates flame and decreases flame temperature.
- It forms smoke and pollutes air.
- Overall regarding burning of coal, the coal with lesser V.M., better is the quality of coal.

3. % Ash

- **Principle**:

Inorganic matter in the coal gets oxidized to form metal oxides and silica, which is non-combustible and left as ash.

- **Method:** The residual coal in the above experiments is heated and burnt in an open crucible at above 750^0c for half an hour. The coal gets burnt. The ash left in crucible is cooled in desiccator and weighed (m_3 gms).

- **Formula:** $$Ash\ \% = \frac{Weight\ of\ Ash}{Weight\ of\ coal} X\ 100$$

$$= \frac{M3}{M} X\ 100$$

Significance of % Ash:

- Ash reduces the calorific value of coal as ash is non-burning part in coal.

- Ash disposal is a problem.

- Hence, lesser the ash %, better is the quality of coal.

4. % fixed carbon

- **Principle**: This is the actual amount of carbon available for combustion, after loss of some carbon in the form of volatile matter.

- **Formula:**

$$\% \text{ Fixed Carbon} = 100 - (\% \text{ Moisture} + \% \text{ Volatile matter} + \% \text{ Ash})$$

Significance of Fixed carbon:

- Carbon is the burning part in coal and higher the FC %, higher the calorific value. Hence a good quality of coal.

Numerical

- **Q.1. 1 g of coal loses 0.03 g at 110^{0}c and then complete combustion leaves 0.08 g of residue. The same coal looses 0.22 g at 950^{0}c. Calculate fixed Carbon %.**

- **Solution: Given**

Weight of moisture in 1 g of coal = 0.03 g

Weight of ash in 1 g of coal = 0.08 g

Weight of (moisture + V.M.) in 1 g of coal = 0.22 gm.

$$Moisture\ \% = \frac{Weight\ of\ moisture}{Weight\ of\ coal} X\ 100$$

$$= \frac{0.03}{1} X\ 100$$

= 3 %

- $Ash\,\% = \frac{Weight\ of\ Ash}{Weight\ of\ coal} X\ 100$

  = $\frac{0.08}{1} X\ 100$

  = 8 %

- $Volatile\ matter\,\% = \frac{Weight\ of\ (moisture+V.M)}{Weight\ of\ coal} X\ 100 - Moisture\,\%$

  = $\frac{0.22}{1} X\ 100 - 3$

  = 22-3

  = 19 %

- Fixed Carbon % = 100- (19+ 3 + 8)

  = 70 %

- **Q.2 a) 2.5 g of coal on heating at 110^{0}c, leaves 2.415 g residue.**
  b)Then the residue on heating at 950^{0}c with cover on crucible gives 1.528 g residue.
  c) The 1.528 g residue on burning gives 0.245 g residue.
  What type of analysis is this? Calculate Percentage results.

- **Solution:** This is the Proximate analysis and Moisture, Volatile Matter, Ash and Fixed Carbon are calculated as below.

Weight of coal = 2.5 g

Weight of moisture = 2.5 - 2.415 = 0.085 g

$$Moisture\,\% = \frac{Weight\ of\ moisture}{Weight\ of\ coal} X\,100$$

$$= \frac{0.085}{2.5} X\,100$$

= 3.4 %

- Weight of Volatile matter = 2.415- 1.528 = 0.887 g
- $$Volatile\ matter\ \% = \frac{Weight\ of\ V.M}{Weight\ of\ coal} X\ 100$$

$$= \frac{0.887}{2.5} X\ 100$$

= 35.5 %

- Weight of coal= 2.5 g
- Weight of ash = 0.245 g
- $$Ash\ \% = \frac{Weight\ of\ Ash}{Weight\ of\ coal} X\ 100$$

$$= \frac{0.245}{2.5} X\ 100$$

= 9.8 %

- Fixed Carbon = 100 - (% Moisture + % Volatile matter + % Ash)

  = 100 – (9.8 + 3.4 + 35.5)

  = 51.3 %

Numerical for practice

- 1. 2.5 g of sample of coal was weighed into a silica crucible. After heating for one hour at 110 ^{0}c, the residue weighed, 2.415 g. The crucible next was covered with a vented lid and strongly heated for exactly 7 min. at 950 ^{o}c. The residue weighed 1.528 gm. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to be 0.245 g. Calculate the percentage results of above analysis.

- 2. 2.4 g of coal sample was weighed in silica crucible. After heating for one hour at 110 ^{0}c, the residue weighed as 2.25 g. the crucible was then covered with a vented lid and strongly heated for exactly 7 min at 950 ^{0}c. The residue weighed as 1.42 g. the crucible was further heated without lid until a constant weight was obtained. The last residue was found to be 0.22 g. calculate the % result of the above analysis.

Ultimate Analysis:

Definition:

The analysis in which % of **C, H, N, S, & O from the coal** can be determined is called Ultimate analysis.

1. C, H in coal:

Principle :-

On combustion, Carbon and Hydrogen present in coal are converted to equivalent amount of CO_2 & H_2O as products of combustion, which get absorbed in KOH tube & $CaCl_2$ tube respectively of known weights & from increase in weights of those tubes, % of C & H can be calculated.

a) $C + O_2 \rightarrow CO_2$

b) $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

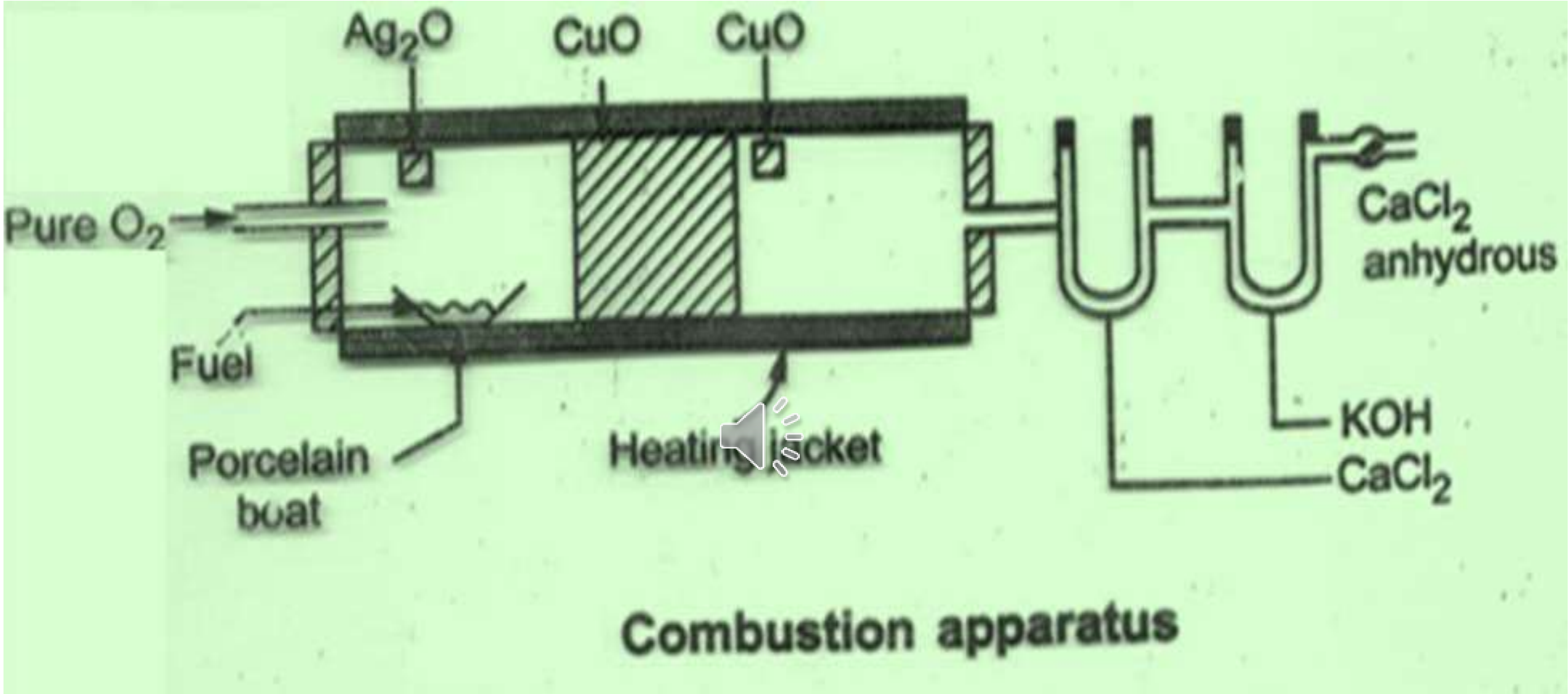


Fig :- Combustion Apparatus

Ref :- Engineering Chemistry by Jain and Jain

Reactions and Formula

1) $C + O_2 \longrightarrow CO_2$
2) $H_2 + ½\, O_2 \longrightarrow H_2O$

$44 \text{ gm } CO_2 \longrightarrow 12 \text{ gm of C}$

$18 \text{ gm } H_2O \longrightarrow 2 \text{ gm of H}$

$$\% C = \frac{Weight\ of\ CO2\ X\ 12\ X\ 100}{Weight\ of\ coal\ sample\ X\ 44}$$

$$\% H = \frac{Weight\ of\ H2O\ formed\ X\ 2\ X\ 100}{Weight\ of\ Coal\ sample\ X\ 18}$$

Significance:

Carbon: Greater the % of carbon in coal, better is the quality and calorific value.

- **Hydrogen** : Most of the hydrogen in coal is in the form of moisture and volatile matter. Only a small percentage of hydrogen is combustible. Hence it decreases C.V. of coal. Smaller the H%, better is the coal quality.

2. Nitrogen in coal: Kjeldahl's method

- **Principle:**

Nitrogen in coal, in presence of conc. H_2SO_4 & K_2SO_4 , converted into equivalent amount of ammonia which is distilled in standard acid solution & from acid consumed by ammonia, percentage of nitrogen in coal can be determined.

$$\mathbf{N\ (Coal)} \xrightarrow[K_2SO_4]{H_2SO_4,\ heat} \mathbf{(NH_4)_2SO_4} \xrightarrow[NaOH]{alkali,\ heat} \mathbf{NH_3} \longrightarrow \mathbf{H_2SO_4\ \&\ Unused\ H_2SO_4 + NaOH}$$

- **Significance**:

Nitrogen does not burn during coal combustion and therefore it has no calorific value. Hence a good quality coal contains negligible N%.

- **Process:** A known weight of powdered and air dried coal is heated with concentrated H_2SO_4 along with K_2SO_4 catalyst in a long necked Kjeldahl's flask.
- After the contents become clear, it is treated with alkali solution in a round bottom flask. The ammonia (basic gas) liberated is passed in known volume of standard acid solution.
- The unused acid is determined by back titration with NaOH solution.

- **Formula:-**

$$\% \text{ N} = \frac{\text{Volume of acid consumed X normality of acid X 1.4}}{\text{Weight of Coal sample}}$$

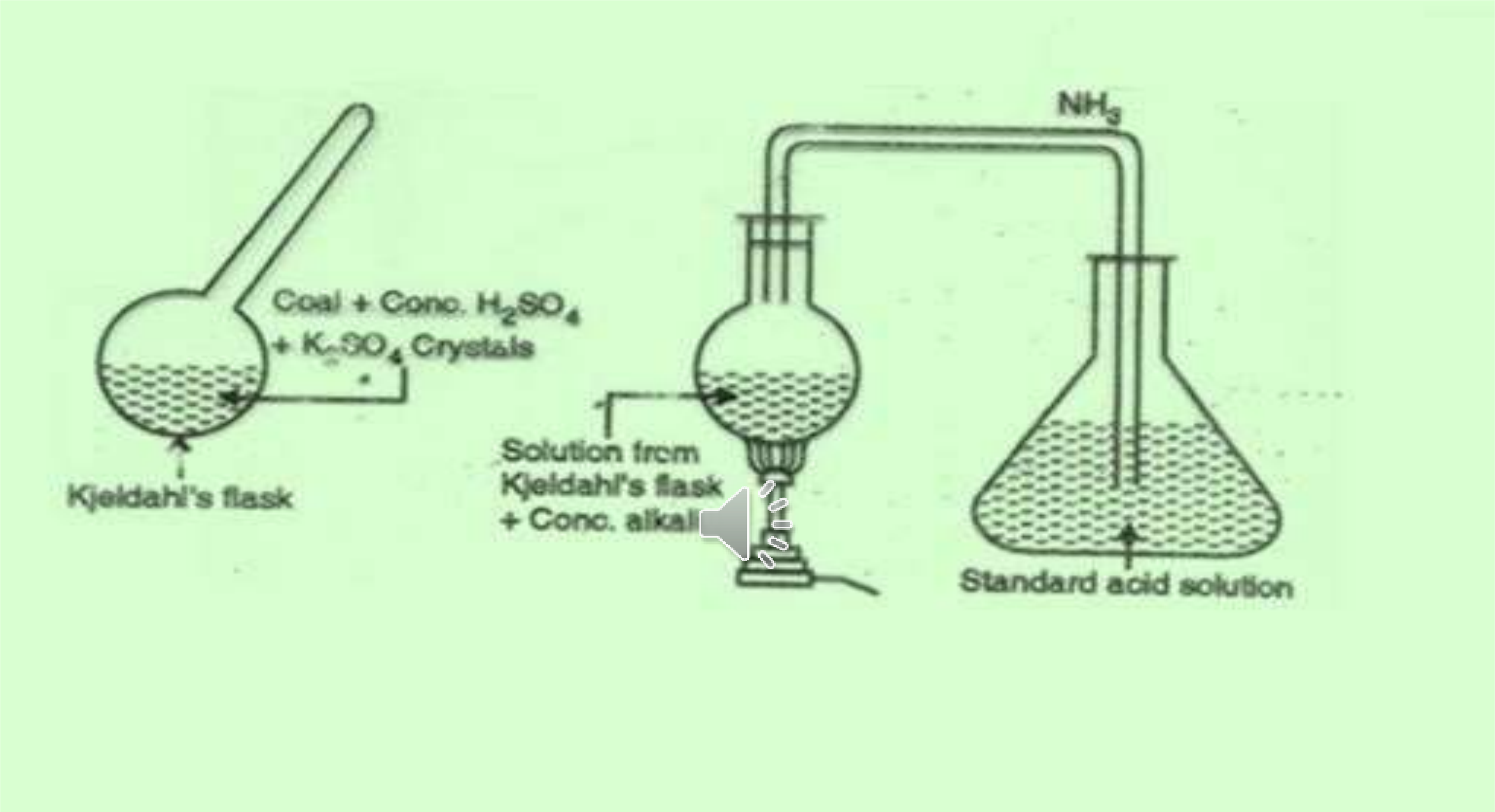


Fig :- <u>Kjeldahl's method</u>

Ref :- Engineering Chemistry by Jain and Jain

Determination of Sulphur

Principle

Sulphur on combustion form sulphur oxides. Washings from Bomb calorimeter have sulphuric acid along with nitric acid. Hence estimation of sulphur is done gravimetrically.

$$S \xrightarrow{O_2} SO_2 \xrightarrow{1/2\ O_2} SO_3 \xrightarrow{H_2O} H_2SO_4 \xrightarrow{BaCl_2} BaSO_4$$

- **Significance**

Although sulphur can burn and increases the calorific value of coal but it causes SOx pollution and causes acid rain, corrosion of metallic equipment. Hence, lower the % of S in coal, better will be the quality.

Process: Take about 10 ml of distilled water in the Bomb pot. Burn the known weight of powdered and air dried coal sample in the Bomb calorimeter experiment.

- Then collect the washings of the Bomb pot in a beaker. Add $BaCl_2$ solution in it.
- Filter the precipitate of $BaSO_4$, dry it and weigh the precipitate of $BaSO_4$, from weight of $BaSO_4$ precipitate, calculate sulphur % as below.
- **Formula**
- $$\% S = \frac{Weight\ of\ BaSO4\ ppt\ X\ 32\ X\ 100}{Weight\ of\ Coal\ sample\ X\ 233}$$

Determination of % O

By knowing the % age of all other constituents in coal % age of oxygen can be calculated as

Formula

$$\% O = 100 - \% (C + H + N + S + Ash)$$

Numerical

- **Q.1 1.6 gm of a coal sample in Kjeldahl's experiment liberated ammonia which was absorbed in 50 ml sulphuric acid. The resultant solution required 14 ml of 0.1 N NaOH for complete neutralization of H_2SO_4 in back titration. The reading for blank titration was 25 ml. Find the % of Nitrogen in coal.**

- **Solution:** Weight of Coal = 1.6 gm

Volume of Sulphuric acid consumed by ammonia = Blank titration reading – Back titration reading

= (25-14) ml of 0.1 N NaOH

= 11 ml of 0.1 N NaOH

$$N\,\% = \frac{Volume\ of\ acid\ consumed\ X\ Normality\ of\ acid\ X\ 1.4}{weight\ of\ Coal\ sample}$$

$$= \frac{11\ X\ 0.1\ X\ 1.4}{1.6}$$

- = 0.963 % of nitrogen in coal

- **Q.2 One gram of coal sample was burnt in Oxygen. Carbon Dioxide was absorbed in KOH and water vapour in $CaCl_2$. The increase in weight of KOH and $CaCl_2$ was 3.157 and 0.504 g respectively. Determine % C and % H in the sample.**

- **Solution: Given**: Weight of CO_2 absorbed in KOH = 3.157 g

Weight of H_2O absorbed in $CaCl_2$ = 0.504 g

Weight of coal burnt = 1.0 g

a) $\% C = \frac{\text{Weight of } CO_2 \text{ X } 12 \text{ X } 100}{Weight\ of\ Coal\ sample\ X\ 44}$

$= \frac{3.157 \text{ X } 12 \text{ X } 100}{1\ X\ 44}$

= 86.1 %

b) $\% H = \frac{\text{Weight of } H_2O \text{ X } 2 \text{ X } 100}{Weight\ of\ Coal\ sample\ X\ 18}$

$= \frac{0.504 \text{ X } 2 \text{ X } 100}{1\ X\ 18}$

= 5.6 %

- **Q.3 Calculate S% in coal when 0.55 g of coal is combusted in Bomb calorimeter. Solution from bomb on treatment with $BaCl_2$ forms 0.025 g $BaSO_4$.**

- **Solution**: **Given**: Coal weight = 0.55 g, Weight of dry $BaSO_4$ ppt = 0.025 g

$$S\,\% = \frac{Weight\ of\ BaSO4\, ppt. X\ 32\ X\ 100}{weight\ of\ Coal\ sample\ X\ 233}$$

$$S\,\% = \frac{0.025\ X\ 32\ X\ 100}{0.55\ X\ 233}$$

= 0.62 % Sulphur in coal

Numericals for Practice

1) 2.1 gm of a coal in Kjeldhal's method release ammonia, which is passed in 50 ml 0.1 N H_2SO_4 solution. To neutralize excess acid 28 ml of 0.1 N NaOH was required. Calculate % N in sample.

(Ans: % N = 1.466)

2) A sample of coal was analyzed as follows:

2.5 gm of coal was weighed into silica crucible. After heating for one hour at 110 ^{0}C, residue weighed 2.415 gm. Then crucible heated at 950^{0}C. The residue weighed 1.528 gm. The crucible was heated till constant weight 0.245 gm was obtained. Calculate % FC. (Ans – % FC = 51.32)

3) 1.5 gm of coal on burning in a combustion tube increases weight of $CaCl_2$ tube by 0.25 gm and KOH tube by 3.05 gm. Find %C and %H.

(%c = 55.45 and %H = 1.85)

4) Calculate S% in coal when 0.75 g of coal is combusted in Bomb calorimeter. Solution from Bomb on treatment with $BaCl_2$, forms 0.025 g $BaSO_4$.

Liquid fuel (Petroleum or crude oil)

- Petroleum or crude oil is a dark greenish brown liquid found in the earth crust at the depth of few meters to 7 km.
- It is the very important source of liquid fuels.

Origin of Petroleum:

- Petroleum is formed from buried debris of plants and animals (organic matter) under favourable conditions.
- Conversion of this material into various hydrocarbons takes place under either influence of radioactive substances or by action of anaerobic bacterial material decomposition in presence of water.
- The anaerobic bacteria, higher temperature, radioactive matter make the degradation of organic matter, in the presence of water to highly alkanes rich matter as crude oil.

Composition of Petroleum oil:

Petroleum oil is composed mainly of various hydrocarbons such as paraffin, cycloparaffins or napthenes, olefins and aromatic together with small amount of organic compounds containing oxygen, nitrogen and sulphur.

The average % composition of elements in crude oil is

1. C= 79.5 to 81%
2. H= 11 to 15 %
3. S= 0.1 to 3.5 %
4. O= 0.1 to 0.9%
5. N= 0.4 to 0.9 %

POWER ALCOHOL

Definition:-

- "When ethyl alcohol is used as fuel in internal combustion engine with petrol for generation of power is called power alcohol.”

OR

- “When ethyl alcohol is blended with petrol as an internal combustion engine is called Power alcohol.”
- Pure ethyl alcohol can not be used as fuel.
- Generally, ethyl alcohol is used as its 5-25% mixture with petrol.

Preparation of Power alcohol:

$$C_{12}H_{22}O_{11} \xrightarrow{\text{Invertase (yeast)}} C_6H_{12}O_6 + C_6H_{12}O_6$$

Sucrose Glucose Fructose

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase in yeast}} 2C_2H_5OH + 2CO_2\uparrow$$

Merits/Advantages of Power Alcohol:

1. It has good antiknocking property and its octane number is 90, while the octane number of petrol is 65. Therefore addition of ethyl alcohol to petrol increases the octane number.

2. Alcohol has property of absorbing any traces of water if present of petrol.

3. Ethyl alcohol contains oxygen atoms which help for complete combustion of power alcohol and polluting emissions of CO, hydrocarbon, particulates are reduced largely.

4. Use of ethyl alcohol in petrol reduces our dependence on foreign countries for petrol and saves foreign currency considerably.

5. Power alcohol is cheaper than petrol.

Disadvantages / Demerits of Power Alcohol

- Ethyl alcohol has C.V. 7000 cal/gm much lower than C.V. of Petrol 11500 cal/gm.
- Ethyl Alcohol has high surface tension and atomisation, especially at lower temperature is difficult causing starting trouble.
- Ethyl alcohol undergo oxidation to form acetic acid which corrodes engine parts.
- Ethyl alcohol obtained by fermentation process directly cannot be mixed with petrol. It has to be dehydrated first.
- As the Ethyl alcohol contains 'O' atoms, the amount of air required for the complete combustion of power alcohol is lesser and therefore carburetor and engine needs to be adjusted or modified, when only ethyl alcohol is used as a fuel.

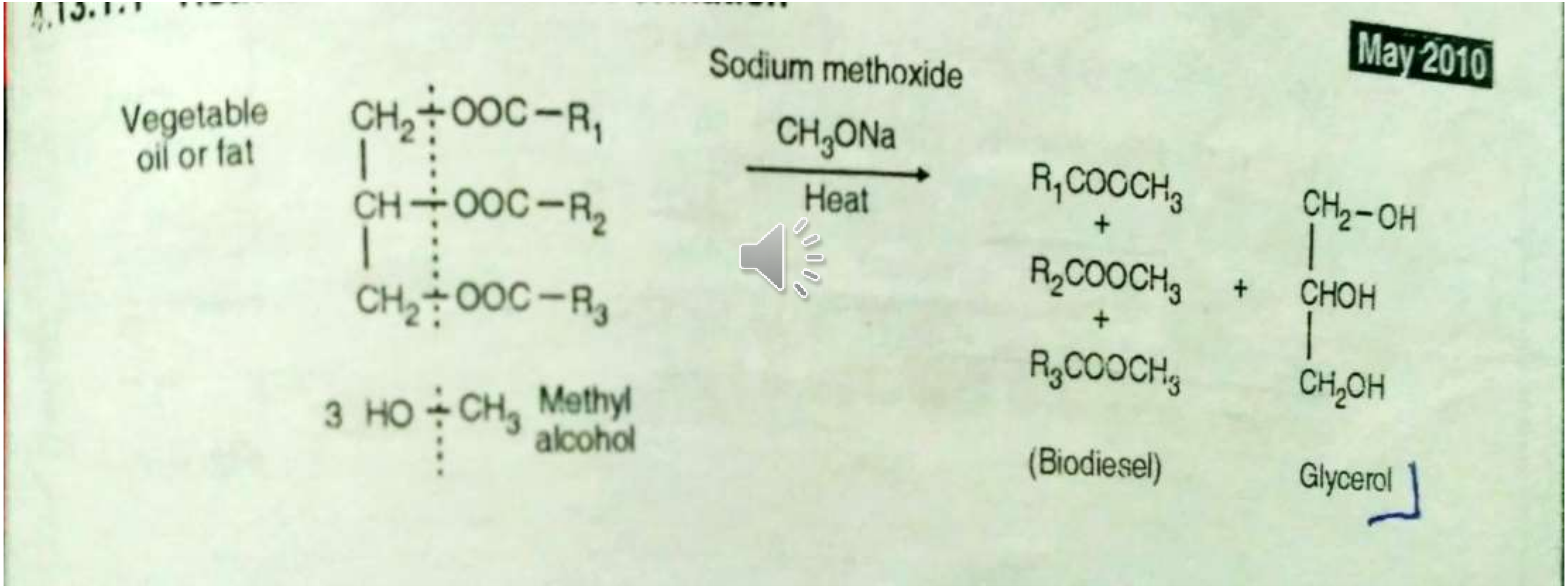
BIODIESEL

- Chemically biodiesel is the methyl esters of long chain carboxylic acids.
- Biodiesel is obtained by transesterification of vegetable oil or animal fats with methyl alcohol using catalyst sodium metal or sodium methoxide.
- Transesterification is the process of converting one ester to another ester.

Preparation of Biodiesel

- Filter the cheap, nonedible or waste vegetable oil or animal fat.
- Heat it at 110^0c with stirring to remove any water from it.
- Add sodium methoxide about 2% by weight to the vegetable oil or animal fat.
- Add methanol about 20% volume to the mixture.
- Reflux the mixture for 30 minutes at 60 -70^0c.
- Cool and mix sufficient water, stir well.
- Separate the water insoluble phase (Biodiesel) from water phase.
- Add antioxidant to the biodiesel to avoid it to become gummy due to oxidation and polymerization.

Reaction


May 2010
Vegetable oil or fat
CH2–OOC–R1
CH–OOC–R2
CH2–OOC–R3
3 HO–CH3 Methyl alcohol
Sodium methoxide
CH3ONa
Heat
R1COOCH3
+
R2COOCH3
+
R3COOCH3
(Biodiesel)
+
CH2–OH
CHOH
CH2OH
Glycerol

Advantages

- Biodiesel is cheaper as it is manufactured from cheap, nonedible or waste oil or animal fats.
- Biodiesel can be used as a good and clean fuel for diesel engines.
- It has high Cetane number 46 to 54 and CV about 40 KJ/gm.
- It is regenerative and environmental friendly.
- It does not give out particulate and CO pollutants.
- It has certain lubricity due to higher oiliness of the esters.
- It provides good market to vegetable oils and reduce our dependence on diesel on foreign countries, saving currency.

Disadvantages /Demerits of Biodiesel

- Cloud and pour point of biodiesel are higher than diesel and can cause problem in fuel flow line. So cannot be used in cold regions.

- Biodiesel may have dissolving action on rubber articles.

- There is shortage of vegetable oils and the starting material if costly, the biodiesel will be costly.

- Biodiesel strongly adhere on metals and can become gummy.

Gaseous fuel

Advantages:

1. Clean to use.
2. They need no excess air for combustion.
3. Rate of combustion is easily controllable.
4. High thermal efficiency.
5. Complete combustion, negligible pollution.

Hydrogen gas

- **<u>Introduction</u>**
- Hydrogen gas is perhaps going to be the most popular fuel in the 21st century.
- It is an increasingly important fuel for industries, vehicles, rockets, missiles, fuel cells etc.
- Hydrogen gas produces a very high temperature blue flame in combustion and no pollutant is emitted in atmosphere by it's combustion.

Properties

- Colourless, odourless and insoluble in water
- Diatomic molecule in which two H atoms are joined by strong covalent bond with bond energy 435.9 KJ/mole
- Density is 0.08987 gm/cc at 0° C and 1 atm. Pressure. It is 14 times lighter than air
- It burns in air forming water and liberates a large amount of energy. This reaction is often explosive
- It reacts with halogen, many metals and $N_{2.}$
- **It is a Clean source of energy, hence it can replace coal and oil as major source of energy in future**
- Liquid hydrogen is used as a fuel in space rockets

Preparation

1) By steam reforming of methane

* It is the process of reacting steam with hydrocarbons in the presence of catalyst at high temperature to obtain hydrogen and carbon oxides.

- 1) $CH_4 + H_20 \xrightarrow{\text{Ni/ FeO cat.}} CO + 3\ H_2$ (at high temperature at 800 ° C and in presence of FeO/ Ni catalyst)

  2) **Shift reaction**

  In shift reaction more steam is mixed with ($CO+H_2$) mixture to get higher % H_2 mixture

- $CO + H_2 + H_20 \xrightarrow{\text{FeO/Cu}} CO_2 + 2H_2$ (at lower temperature 400 °C and using FeO/Cu catalyst)

Then the mixture (CO_2+H_2) is compressed and cooled to get liquid CO_2 and H_2

Gas. The alternate way to remove CO_2 from the mixture is scrubbing the (CO_2+H_2) mixture with alkali solution or a solvent which can be recycled after evaporating out CO_2.

95-98% purity will be obtained.

2) By steam reforming of coke

Coal on heating with superheated steam produces the water gas which is then reacted with more steam to produce additional hydrogen by shift reaction.

- $C + H_2O \xrightarrow{\textbf{Ni/ FeO cat.}} CO + H_2$ (at high temperature at 700 ° C and in presence of Ni/FeO catalyst)

  (Water gas)

Water gas shift reaction

- $CO + H_2 \xrightarrow{\textbf{FeO},\ H2O,\ 400\ °C} CO_2 + 2\ H_2$ (at lower temperature 400 °C and using Fe catalyst)

The H_2 obtained in this process is less and it is associated with impurity of H_2S, SO_2. Hence this process is less preferred, although very cheap method.

Hydrogen as a future fuel

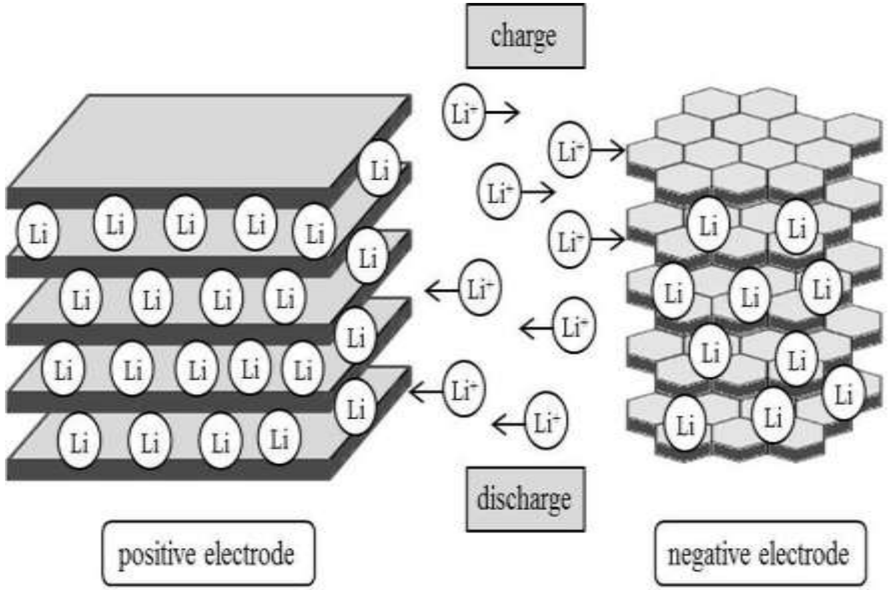
- Hydrogen gas is going to be the most popular fuel in the 21st century. It is an increasingly important fuel for industries, vehicles, rockets, fuel cells etc
- Hydrogen gas produces a very high temperature blue flame in combustion and no pollutant is emitted in atmosphere by it's combustion. It has high calorific value.
- After shortage of crude oil fractions in the near future, hydrogen is likely to have increased importance as fuel for industrial and transportation purposes.
- It can be used in fuel cell or internal combustion engine to power vehicles or electric devices.
- It can provide motive power for liquid propellant rockets, cars, boats and aeroplanes.
- Combustion engines in commercial vehicles have been converted to run on a hydrogen-diesel mix where up to 70% of emissions have been reduced during normal driving conditions.
- Minor modifications are needed to the engines a s well as the addition of hydrogen tanks at a compression of 350 bars. It can also be used for heavy duty vehicles where it is expected to run 300km/17kg which means an efficiency better than a standard diesel engine.

A Lithium-ion battery

A Lithium-ion battery is defined as a rechargeable battery that utilizes lithium ions moving between electrodes during charging and discharging processes.

Lithium batteries possess metallic lithium as an anode material. They are quite unique when compared to other batteries because of their high cost per unit and high energy density.

A lithium battery operates on the principle of intercalation and deintercalation of ions from a positive electrode material and a negative electrode material, with the most common type being the Lithium-ion battery.



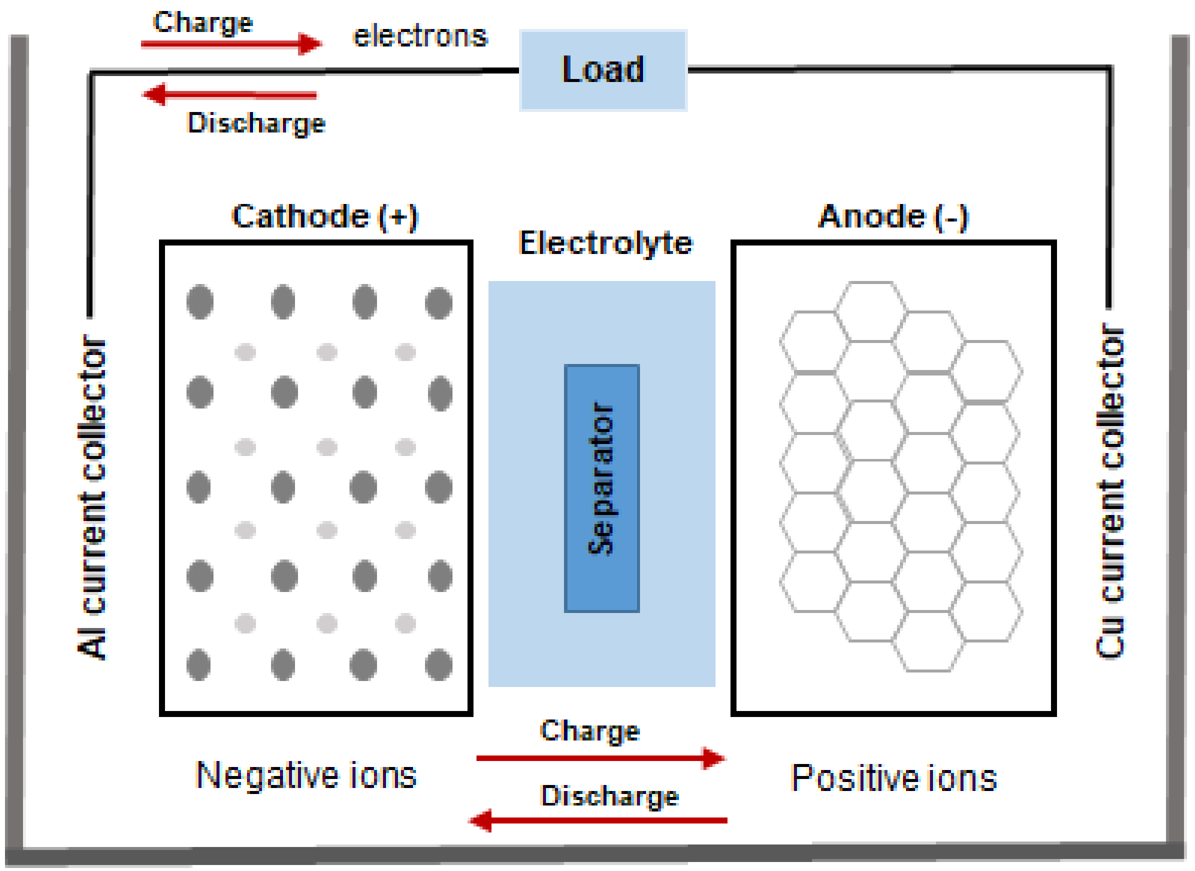


Fig: LI-ION BATEERY

Construction:

Electrodes: The positively and negatively charged ends of a cell. Attached to the current collectors
Anode: The negative electrode
Cathode: The positive electrode
Electrolyte: A liquid or gel that conducts electricity
Current collectors: Conductive foils at each electrode of the battery that are connected to the terminals of the cell. The cell terminals transmit the electric current between the battery, the device and the energy source that powers the battery
Separator: A porous polymeric film that separates the electrodes while enabling the exchange of lithium ions from one side to the other

Working

In a lithium-ion battery, lithium ions (Li+) move between the cathode and anode internally.. This migration is the reason the battery powers the device—because it creates the electrical current.

While the battery is discharging, the anode releases lithium ions to the cathode, generating a flow of electrons that helps to power the relevant device.

When the battery is charging, the opposite occurs: lithium ions are released by the cathode and received by the anode.

At the cathode

$CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$ (**Reduction)**

At the anode

$LiC_6 \rightarrow C_6 + Li^+ + e^-$ (**Oxidation)**

Here is the full reaction (left to right = discharging, right to left = charging):

$LiC_6 + CoO_2 \rightleftarrows C_6 + LiCoO_2$

Lithium Battery Advantages
Safety. Inherently safe chemistry.
Drop-in Replacement. Available in standard industry sizes.
Lightweight. 50-60% less weight than lead-acid equivalent.
Longest Life.
More Usable Capacity.
Constant Power.
Temperature Tolerant.
Charging - Fast & Safe.

Disadvantages of Lithium-ion Battery
1 Relatively Higher Cost. The utility of rare metals in lithium-ion batteries increases the cost of raw materials and refining processes. ...
2 Slightly Heavier Weight. .
3 Repair and Maintenance Challenges. ...
3.4 Safety concerns.

Applications of Li-ion Batteries

• The Li-ion batteries are used in cameras, calculators, smart phones and most of the consumer electronics device.

• They are used in cardiac pacemakers and other implantable device.

• Electric vehicles: Because of their light weight Li-ion batteries are used for propelling a wide range of electric vehicles such as aircraft, electric cars, Pedelecs, hybrid vehicles, advanced electric wheelchairs, radio-controlled models, model aircraft and the Mars Curiosity rover.

• Power tools: Li-ion batteries are used in tools such as cordless drills, sanders, saws and a variety of garden equipment including whipper-snippers and hedge trimmers

Thank you